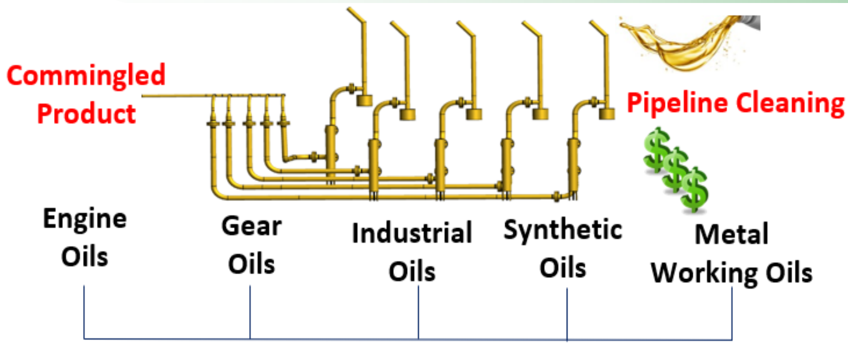
Experimental & Computational Studies to Optimize Petroleum Pipeline Cleaning Operations

Rowan University through support from the U.S. Environmental Protection Agency Pollution Prevention Program received a grant to improve cleaning efficiency in multiproduct petroleum pipelines. The proposed solution integrated experimental analysis, process modeling, optimization and management of change to improve operational efficiency by ~**67%**.

Fabricated Pilot Plant at Rowan



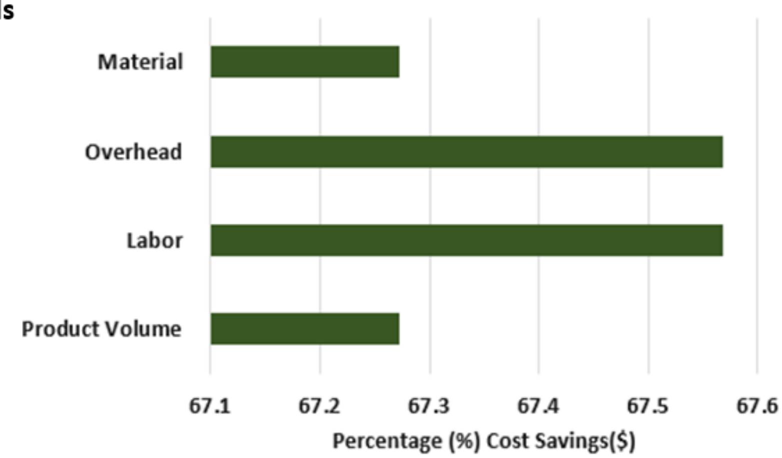
Background

Cleaning of pipelines between product changeovers is crucial for maintaining the product quality and purity in multi-product pipeline systems. Existing cleaning methods are inefficient and lead to formation of low value mixed oils. Our goal is to optimize these cleaning operations and minimize the oil downgrade.

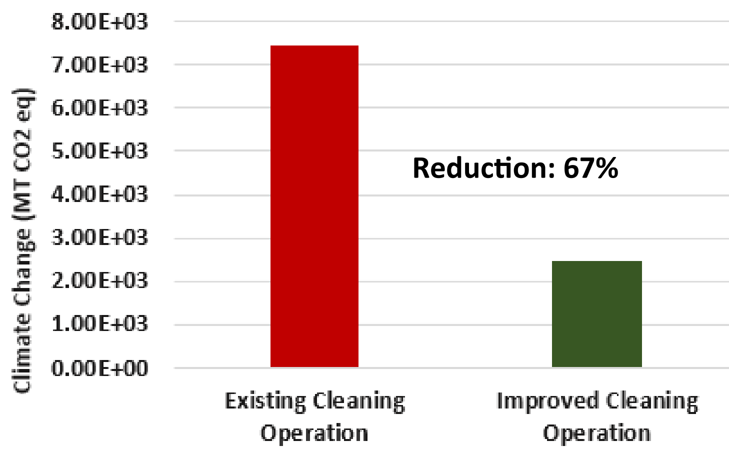
Solution Strategy

- **Operational Review:** Conducted thorough analysis of current pipeline operations to pinpoint areas for improvement
- **Data Analysis**: Explored plant data through statistical analysis for in-depth understanding and insights
- **Alternative Pipeline Cleaning Potential:** Explored several alternatives to improve cleaning operations
- **Feasibility Assessment:** Assessed the drawbacks and practicality of the alternative methods explored
- **Selection of Candidate Solution:** Chose procedural enhancement as the best alternative solution
- **Benchtop Rig Development**: Built a benchtop rig to test the efficacy of the procedural enhancement
- **Management of Change Operations:** Implemented the improved procedure at the plant
- **Pilot Plant Development:** Fabricated a pilot plant to refine the improved procedure and model the cleaning operations

Estimated Cost Savings*



Environmental Impact Assessment*



Total Projected Cost Savings = **$ 88,119.00**

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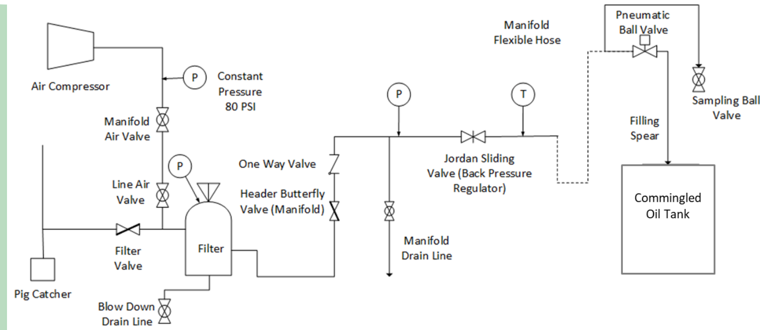
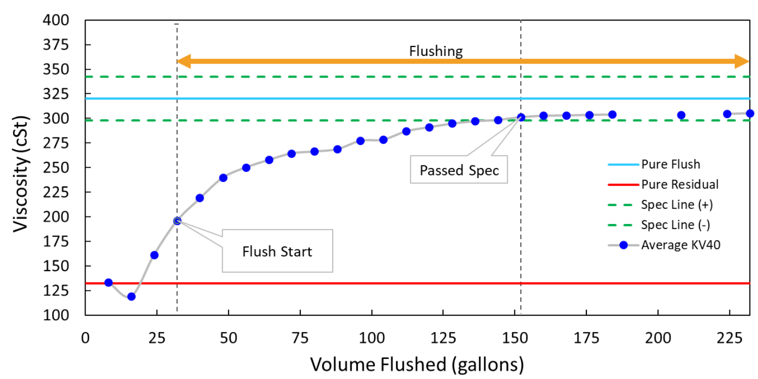
Progression to Pollution Prevention Solution

- **Overview Meeting:** Initial meetings were conducted with the company to identify the pollution prevention problem.
- **Plant Operation Visits**: In-person visits to the industrial facility to observe the overall process. At these visits the overall process of oil flow from refinery to final packaging.
- **Scoping of problem:** Through meetings and discussions with industry operators and engineers the opportunity for pollution prevention was identified to be the process equipment after the pig collection device.
- **Drum Fill:** A detailed process flow diagram was constructed of the equipment, fittings and piping of the drum fill line.
- **Observation of operation of Drum fill line.** Students observed the operators running the drum line. This resulted in a plan to obtain samples from the process that could be analyzed in the laboratory.
- **Industrial Runs**: Samples taken every 10 seconds by plant operators with students observing runs.
- **Characterization of Flushing Curve** Viscosity of samples were analyzed to obtain a step curve of flushing from the previous oil to the new oil.
- **Bench Top Apparatus** A laboratory apparatus was built to scope out possible changes to the operating procedure of the drum fill line.
- **Air Blowing Experiments:** Built an industrial scaled down pilot plant to refine the improved procedure and model the cleaning operations

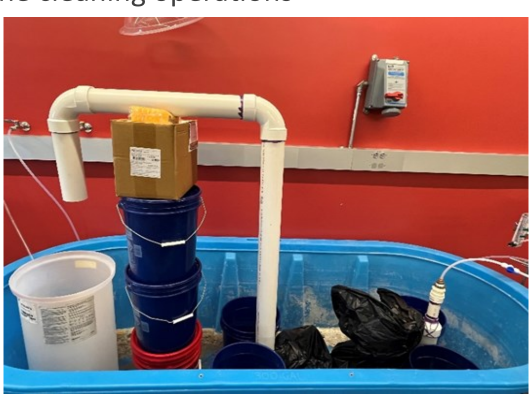


Schematic of the Drum Filling Line

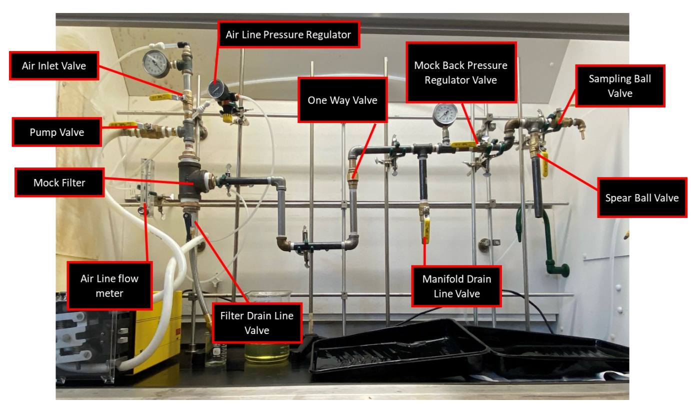
Oil samples collected every 10 sec of flushing



Sample flushing results during changeover



Testing of 3” diameter PVC pipe for Air Blowing Experiment



Benchtop setup of the lube oil packaging line

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**Based on Projected Operations*

Link to journal and peer-reviewed conference publications:

1. Jerpoth, S. S.; Hesketh, R.; Slater, C. S.; Savelski, M. J.; Yenkie, K. M. Strategic Optimization of the Flushing Operations in Lubricant Manufacturing and Packaging Facilities. *ACS Omega* **2023**, *8* (41), 38288–38300. https://doi.org/10.1021/acsomega.3c04668.
2. Jerpoth, S. S.; Hesketh, R.; Slater, C. S.; Curtis, S.; Fracchiolla, M.; Theuma, D.; Yenkie, K. M.* Hands-on Experience in Solving Real-World Problems via a Unique Student-Faculty-Industry Collaboration Program. 2023 ASEE Annual Conference, Baltimore, MD. https://strategy.asee.org/43334
3. Jerpoth, S.S.; Hesketh, R.P.; Slater, C. S.; Savelski, M.J.; McClernan, R.; Yenkie, K.M. Application of Discretized Non-Linear Programming to Minimize Mixed Oil Formation in Flushing Operations of Lubricant Pipelines. Proceedings of the *Foundations of Computer-Aided Process Operations and Chemical Process Control (FOCAPO/CPC)*, 2023
4. Jerpoth, S.S.; Hesketh, R.P.; Slater, C. S.; Savelski, M.J.; Yenkie, K.M. Computational Modeling of Lube-Oil Flows in Pipelines to Study the Efficacy of Flushing Operations. Proceedings of the 14th International Symposium on Process Systems Engineering (PSE), 2022.

Link to standard operation procedure (SOP) for improved cleaning GitHub: https://github.com/kmygroup/EPA_P2_Optimization-of-Pipeline-Processes